



Master Your Problem Solving Skills.

Brief overview

1. An online coding platform supporting multiple languages with real-time compilation and verdict system.

2. A 1v1 competitive mode with room creation, random problem assignment, live results, and an Elo-based rating system with leaderboard.

3. Implement problem solving with test cases, along with separate admin and user panels for management and tracking.

4. Provide user features including friend-based chat, a blog section for algorithms and discussions, and detailed profile dashboards with heatmaps, ratings, and pie chart analytics.

List of Contents

01. Homepage

02. Login/ Register

03. Problems

04. Compete

05. Blogs

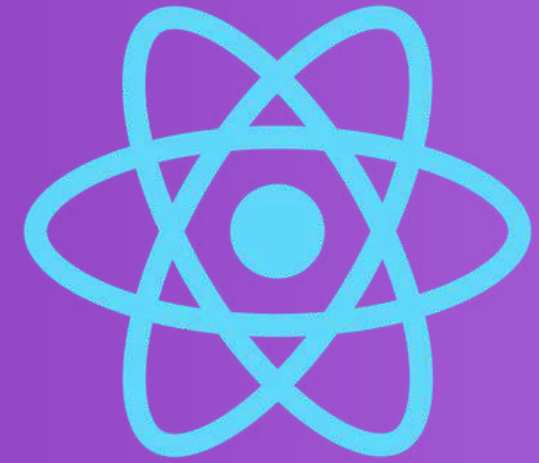
06. Submissions

07. Users

08. Messages

Tech Stack

Frontend

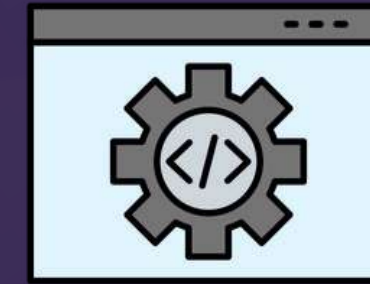


React.js

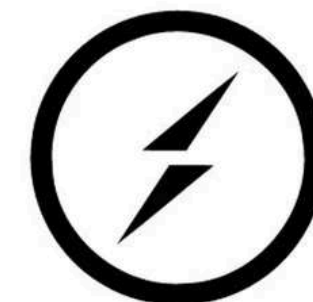


Database

Mongodb



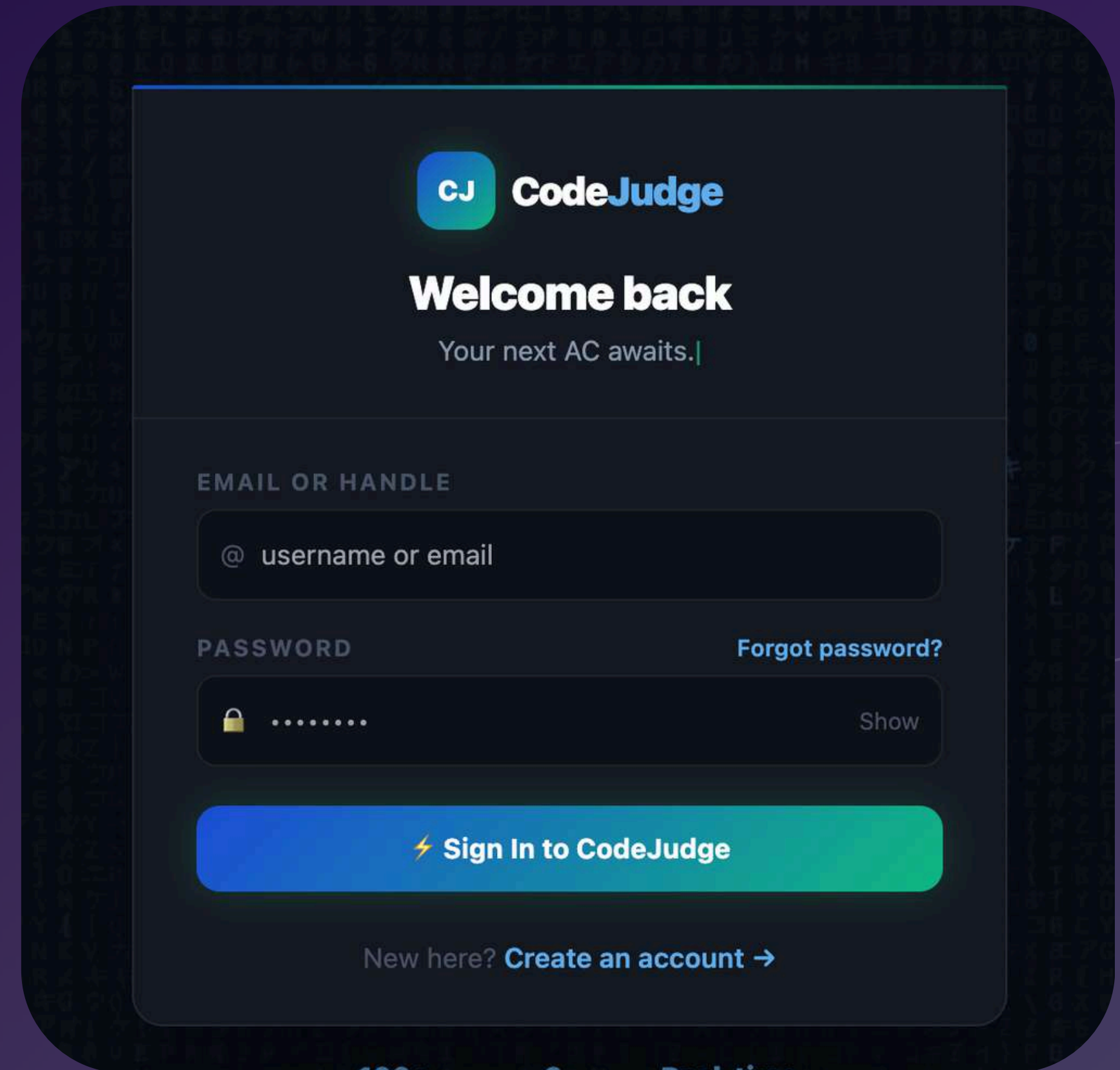
Backend



socket.io

Auth(Login)

- Login using email or username
- Secure password authentication with hashing
- Forgot password option available
- OTP verification via email for password reset
- Set and update new password securely

A screenshot of the CodeJudge login page. The page has a dark theme with a blue and green gradient. At the top, the CodeJudge logo is displayed. Below the logo, the text "Welcome back" is shown, followed by "Your next AC awaits." in a smaller font. The login form consists of two main sections: "EMAIL OR HANDLE" and "PASSWORD". The "EMAIL OR HANDLE" section has a text input field with a placeholder "@ username or email". The "PASSWORD" section has a password input field with a lock icon and a "Show" button. To the right of the password field is a link "Forgot password?". Below the password field is a large blue button with a lightning bolt icon and the text "Sign In to CodeJudge". At the bottom, there is a link "New here? Create an account →".

CodeJudge

Welcome back
Your next AC awaits.

EMAIL OR HANDLE

@ username or email

PASSWORD [Forgot password?](#)

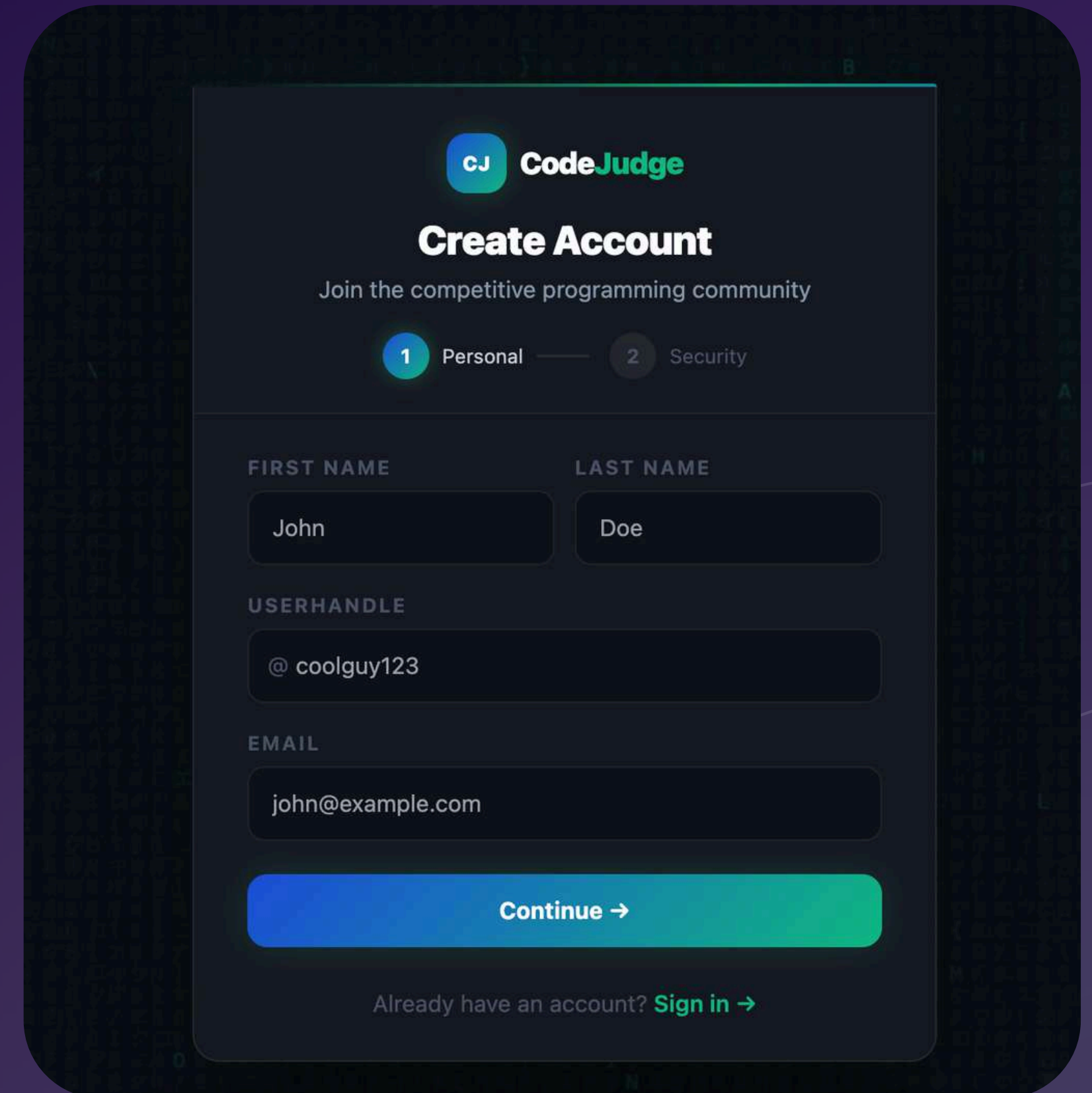
..... Show

⚡ Sign In to CodeJudge

New here? [Create an account →](#)

Auth(Register)

- User registration with personal details
(name, handle, email)
- Secure authentication with validation
- Step-based signup process
(Personal → Security)
- Unique username (handle) and email for each user



The image shows a mobile app interface for creating an account on CodeJudge. The app has a dark theme with a background of faint, colorful code characters. At the top, the CodeJudge logo is displayed. Below it, the title 'Create Account' is centered, followed by the subtitle 'Join the competitive programming community'. A progress indicator shows two steps: '1 Personal' (active) and '2 Security'. The form contains three input fields: 'FIRST NAME' with the value 'John', 'LAST NAME' with the value 'Doe', and 'USERHANDLE' with the value '@ coolguy123'. Below these is an 'EMAIL' field with the value 'john@example.com'. A large blue button labeled 'Continue →' is positioned below the email field. At the bottom, there is a link that says 'Already have an account? Sign in →'.

CodeJudge

Create Account

Join the competitive programming community

1 Personal — 2 Security

FIRST NAME LAST NAME

John Doe

USERHANDLE

@ coolguy123

EMAIL

john@example.com

Continue →

Already have an account? [Sign in →](#)

Platform -features

PLATFORM FEATURES

Everything you need to **level up**

From competitive battles to community blogs — explore all the tools designed to make you a better coder.



1v1 Compete

Challenge friends to real-time coding battles. First to AC wins!

Explore →



Problem Set

100+ curated problems across DSA, algorithms, and all difficulty levels.

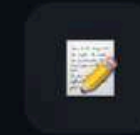
Explore →



Rating System

Earn rating points, climb tiers from Bronze to Champion. Track your progress.

Explore →



Blogs

Read & write technical articles. Share knowledge with the community.

Explore →



Submissions

Track all your submissions with detailed verdicts, time & memory stats.

Explore →



Community

Connect with coders, add friends, and view public profiles.

Explore →



Real-time Judge

Instant verdicts with execution time, memory usage, and detailed errors.

Explore →



Leaderboard

Compete globally. See top coders by rating, submissions, and acceptance.

Explore →

Homepage : Level Up Your Coding Journey



Master Your Competitive Coding Skills.

Solve problems, compete with peers, and improve your algorithms on [CodeJudge](#).

[Get Started for Free](#)

[View Submissions →](#)

```
44
45 int main() {
46     ios_base::sync_with_stdio(false); cin.tie(NULL);
47     int n, q; cin >> n >> q;
48     int arr[n + 1];
49     for (int i = 1; i <= n; ++i) cin >> arr[i];
50     build(1, 1, n, arr);
51
52     while (q--) {
53         int type; cin >> type;
54         if (type == 1) { // point update
55             int idx, val; cin >> idx >> val;
56             update(1, 1, n, idx, val);
57         } else { // range max query
58             int l, r; cin >> l >> r;
59             cout << query(1, 1, n, l, r) << "\n";
60         }
61     }
62 }
```

✓ Accepted Time: 48ms · Memory: 6MB

Practice problem sets in multiple languages, connect with peers, and learn through blogs—all in one platform.

Problem set

Problem Set

11 problems • 6 solved

+ Create Problem

🔍 Search by name or PID...

All 11

Easy 4

Medium 6

Hard 1

🔒 Hide Tags

Filter by Tag:

All (11)

array (1)

string (1)

dp (3)

greedy (3)

math (1)

arrays (1)

constructive algorithms (1)

strings (1)

implementation (1)

STATUS		TITLE	TAGS	DIFFICULTY	EDIT	DEL
✓	1	Two Sum	dp	Easy	Edit	Del
✓	2	Special Elements	—	Medium	Edit	Del
	3	3	—	Easy	Edit	Del
✓	4	Minimum Removals to Achieve Target XOR	arrays	Medium	Edit	Del
	5	The Arrangement	constructive algorithms greedy	Medium	Edit	Del
	6	Wildcard Matching	dp string greedy	Medium	Edit	Del

- Admin can create, edit, and delete problems
- Users can solve problems and track progress
- Search problems by name or problem ID
- Filter problems using tags (DP, Greedy, Arrays, etc.)
- Difficulty - based filtering (Easy, Medium, Hard)
- Toggle tags on/off for a cleaner view
- Organized problem list with status and details

Problem Description

- Clean and structured problem descriptions
- MathJax integration for proper mathematical formatting
- Well-formatted input and output sections
- Standardized test cases with clear examples
- Includes problem ID, description, constraints, input format, and output specification in a structured layout

8 Apple Division Easy

[Description](#) [Submissions](#)

DESCRIPTION

There are n apples with known weights. Your task is to divide the apples into two groups so that the difference between the weights of the groups is minimal.

INPUT

The first input line has an integer n : the number of apples.
The next line has n integers $p_1, p_2, \dots, p_n$: the weight of each apple.

OUTPUT

Print one integer: the minimum difference between the weights of the groups.

CONSTRAINTS

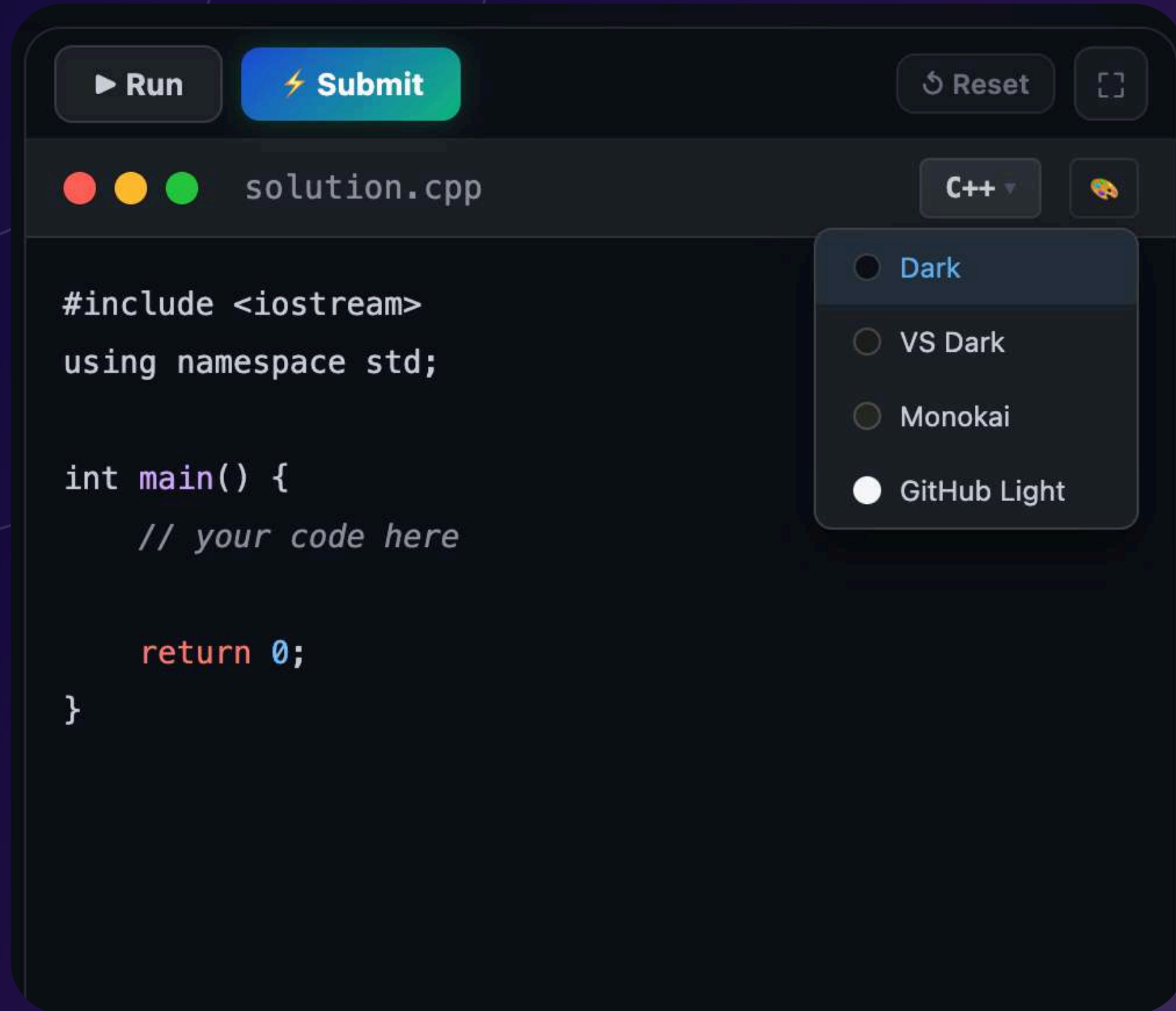
$1 \leq n \leq 20$
 $1 \leq p_i \leq 10^9$

⌚ Time Limit: 2s

SAMPLE

INPUT	OUTPUT
10 952 775 292 702 859 719 65	1

Compiler



- Supports 3 programming languages (C++, Python, Java)
- Multiple coding themes (Dark, VS Dark, Monokai, GitHub Light)
- Auto-save mode to prevent code loss
- Smooth and responsive code editor experience
- Easy run & submit functionality for quick testing

Compiler-backend

- Receive : The user clicks "Run," and the website sends their code and input to your Node.js server.
- Save : The server takes that data and creates temporary source code and input files (using generateFile.js).
- Compile/Prepare : Based on the language, the server uses scripts like executeCpp.js to compile the code (if necessary) into a runnable program.
- Execute : The server runs the program in the terminal, feeds it the input file, and records whatever it prints out (or any errors it throws).
- Respond : The server bundles up those terminal results and sends them back to the website to be displayed on the user's screen.

Admin - Panel

- Admin can create, edit, and delete problems
- Add detailed descriptions, inputs, outputs, and constraints
- Upload and manage test cases for evaluation
- Assign difficulty levels (Easy, Medium, Hard)
- Add tags for better categorization and search

CODEJUDGE · ADMIN

Create Problem

Fill in all fields to publish a new problem.

● Problem Details

Problem ID Unique identifier

e.g. A001

Problem Name

Enter the title

Description Supports plain text

Describe the problem in detail...

Input Format

Explain the input format

Output Format

Explain the expected output

Constraints

e.g. $1 \leq N \leq 10^5$

Tags Press Enter or click Add

e.g. arrays, dp, greedy + Add

Time Limit in seconds

e.g. 2

Difficulty Level

☒ Easy ☐ Medium ☐ Hard

← Cancel + Create Problem

TestCases

- Admin can create test cases for each problem
- Edit existing test cases easily
- Delete test cases when needed
- Provide custom input and expected output
- Manage multiple test cases efficiently for evaluation

Testcases

10 testcases for this problem

[+ Add Testcase](#)

TESTCASE NAME	EDIT	DEL
01 1	Edit	Del
02 10	Edit	Del
03 2	Edit	Del
04 3	Edit	Del
05 4	Edit	Del
06 5	Edit	Del
07 6	Edit	Del

CODEJUDGE · ADMIN

Create Testcase

Adding testcase for problem 9

Testcase Details

Test Name

Used for identification

e.g. Sample Test 1

Input

Raw input data

Expected Output

Exact expected output

e.g.
2
3 5

e.g.
8

[← Cancel](#)

[+ Create Testcase](#)

Submission-page

- See personal submissions and track performance
 - View code of any submission for learning and comparison
 - Admin can delete inappropriate submissions
-
- Track performance with execution time and details
 - Real-time status (Accepted, Wrong Answer, etc.)
 - Built using React.js (frontend) and Node.js (backend)

CODEJUDGE

All Submissions

123 submissions

All Friends Mine

#	TIME	USER	PID	PROBLEM	LANG	STATUS	EXEC TIME	CODE
123	22 Apr 2026 19:41	cdxdx	8	Apple Division	C++	✓ Accepted	76 ms	View
122	22 Apr 2026 19:31	cdxdx	9	Barnicle	C++	✓ Accepted	8 ms	View
121	22 Apr 2026 19:25	amitak17	1	Two Sum	C++	✓ Accepted	6 ms	View
120	22 Apr 2026 19:25	cdxdx	1	Two Sum	C++	✓ Accepted	6 ms	View
119	22 Apr 2026 19:25	amitak17	1	Two Sum	C++	✗ Wrong on Test 1	—	View
118	22 Apr 2026 19:22	cdxdx	222	2	C++	✓ Accepted	6 ms	View
117	22 Apr 2026 19:22	amitak17	1	Two Sum	C++	✗ Wrong on Test 1	—	View
116	22 Apr 2026 19:21	amitak17	1	Two Sum	C++	✓ Accepted	6 ms	View
115	22 Apr 2026 17:46	sidhant08	9	Barnicle	Python	✗ Wrong on Test 1	—	View
114	22 Apr 2026 17:45	sidhant08	9	Barnicle	C++	✓ Accepted	11 ms	View

User - section

CODEJUDGE

Users

18 users

Search by handle or name...

HANDLE	NAME	RATING	SUBMISSIONS	ACCEPTED	ROLE	EDIT	SWITCH	DEL
 sidhant08	Sidhant Nirupam	 1315	106	6	 Admin	Edit	Switch	Del
 sidhant	Sidhant 2	 965	5	3	User	Edit	Switch	Del
 Sidhant_8	Sidhant 3	 160	1	1	User	Edit	Switch	Del
 sidhant_2	Sidhant Nirupam	 0	0	0	User	Edit	Switch	Del
 sidhant_8	Sidhant Nirupam	 0	0	0	User	Edit	Switch	Del
 cool	Sidhant Nirupam	 0	0	0	User	Edit	Switch	Del
 vaibhav	Vaibhav yadav	 34	0	0	User	Edit	Switch	Del
 harendra bhai	Harendra bhai	 28	1	1	User	Edit	Switch	Del
 sidhant22	Sidhant Nirupam	 45	0	0	User	Edit	Switch	Del
 sidhant002	sidhant 3	 120	0	0	User	Edit	Switch	Del

- View users with details (handle, name, rating, submissions, accepted)
- Sort users based on different columns

Edit -User

- Admin can edit and delete users
- Change user roles (Admin/User) easily
- Admin can change handle, name, Email of the User
- Manage user data through a clean and intuitive interface

CODEJUDGE · ADMIN

Edit User

Editing @sidhant

User Details

New Handle

Changing this updates the login handle

@sidhant

First Name

Sidhant

Last Name

2

Email

sidhantmdp@gmail.com

← Cancel

✓ Update User

Leaderboard

Desktop View

[Home](#) [Problems](#) [Compete](#) [Blogs](#) [Submissions](#) [Users](#) [Login](#) [Register](#)

**Leaderboard**
Top 100 players ranked by rating

Search player...

**sidhant08**
1315

2

**cdxdx**
2000

1

**sidhant**
965

3

RANK	PLAYER	RATING	BATTLES	WIN RATE
1	  cdxdx cdxdx	 2000	1	0%

Mobile View

**Leaderboard**
Top 100 players ranked by rating

Search player...

**sidhant08**
1315

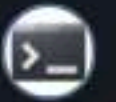
2

**cdxdx**
2000

1

**sidhant**
965

3

  **cdxdx**
1 battles - 0% WR  2000

  **sidhant08**
30 battles - 87% WR  1315

  **sidhant**
24 battles - 8% WR  965

1v1 Compete – Real Time Coding Battles

Battle Entry Options

- Users can create a new room with custom settings
- Or join an existing room using a Room ID

Live Room Access

- Displays public battle rooms available to join
- Enables quick matchmaking with other users

Performance Overview

- Shows current rating and rank level
- Tracks wins, total games, and win rate
- Displays current and best streaks

User Experience

- Clean and intuitive layout for easy navigation
- Designed for fast access to battles with minimal steps

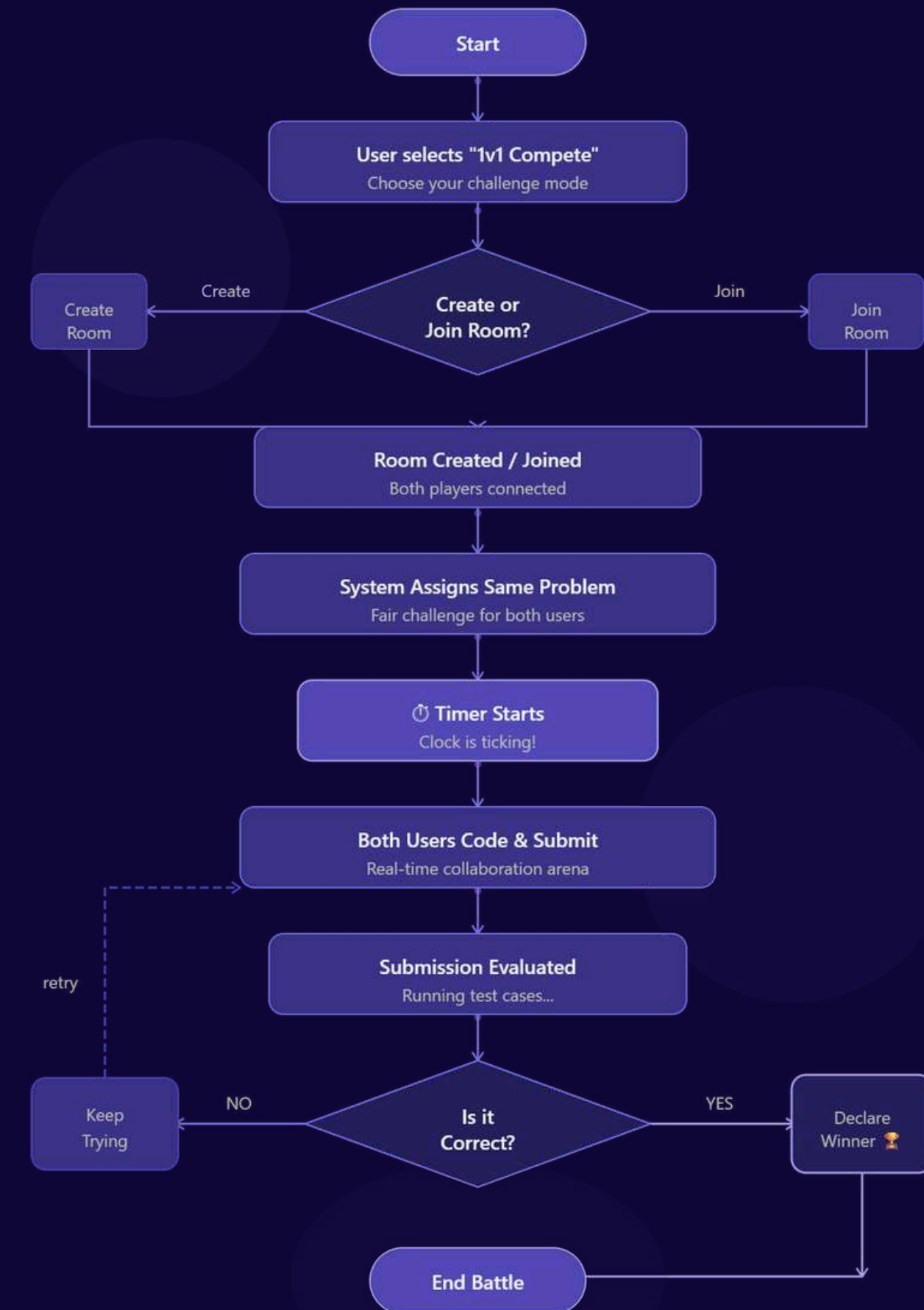


System Purpose

- Central hub for managing 1v1 battles and user stats
- Ensures smooth transition from entry to gameplay

How 1v1 Compete Works

- Start: User selects 1v1 Compete
- Room: Create (generate ID) / Join (enter ID + password)
- Wait: System shows “Waiting for Opponent”
- Sync: Both players connected → battle initialized
- Problem: Same question assigned to both users
- Timer: Countdown starts simultaneously
- Code: Users write & submit solutions
- Evaluate: Submissions run on test cases
- Decision: If correct, declare winner; otherwise, retry
- Winner: First correct submission wins
- End: Battle finishes



Create BattleRoom- Configuration Panel

- The user selects the difficulty level for the battle.
 - The battle duration is set using a slider.
- An optional password can be added for a private room.
- If left empty, the room becomes publicly accessible.
- The user can customize match settings before starting.
- The interface allows quick and easy room creation.
- One-click Create Room to start battle .

The screenshot shows a mobile application interface for creating a battle room. At the top, it says 'CODEJUDGE ARENA' and 'Create Battle Room' with a close button (X). Below this, there are three sections: 'DIFFICULTY' with buttons for 'Easy', 'Medium', 'Hard', and 'Random' (which is highlighted); 'DURATION' with a slider set to '30 min' and labels for '5m', '15m', '30m', '45m', and '60m'; and 'PASSWORD (optional)' with a text input field containing 'Leave empty for public room'. At the bottom, there are three buttons: 'Random' (with a flag icon), '30 min' (with a clock icon), and 'Public' (with a globe icon). At the very bottom, there are two large buttons: 'Cancel' and 'Create Room' (with a rocket icon).

Match Result & Rating Calculation System



- Two players are matched in real time and given the same coding problem with a shared timer starting simultaneously.
- Each player independently writes, tests, and submits code, with evaluation done against hidden test cases.
- The battle can end early if one player submits the first correct solution, or after both submissions are evaluated.
- Ratings are updated using the core Elo update equation: $R' = R + K \cdot (S - E)$
- ELO formula uses adaptive K-factor:

$$K = \begin{cases} 80 & \text{battles} < 4 \\ 40 & 5 \leq \text{battles} < 20 \\ 20 & \text{battles} \geq 20 \end{cases}$$



FIPML9

Barnicle

Hard

29:15

cdxdx

9 Barnicle

Hard

DESCRIPTION

Barney is standing in a bar and starrng at a pretty girl. He wants to shoot her with his heart arrow but he needs to know the distance between him and the girl to make his shot accurate.

Barney asked the bar tender Carl about this distance value, but Carl was so busy talking to the customers so he wrote the distance value (it's a real number) on a napkin. The problem is that he wrote it in scientific notation.

The scientific notation of some real number x is the notation of form AeB , where:

- A is a real number between 0 and 9
- B is a non-negative integer
- $x = A \times 10^B$

Barney doesn't know anything about scientific notation. Your task is to convert the scientific notation to usual decimal representation with minimal number of digits after the decimal point (and no decimal point if it is an integer).

Understanding the conversion:

When you multiply by 10^B , you shift the decimal point B places to the right.

Run

Submit

Forfeit

solution.cpp

C++

```
#include<bits/stdc++.h>

using namespace std;

#define fast ios_base::sync_with_stdio(false); cin.tie(NULL)
#define in long long int
#define vi vector<in>
#define vii vector<vector<in>>
```

CUSTOM INPUT

8.77056e6

OUTPUT

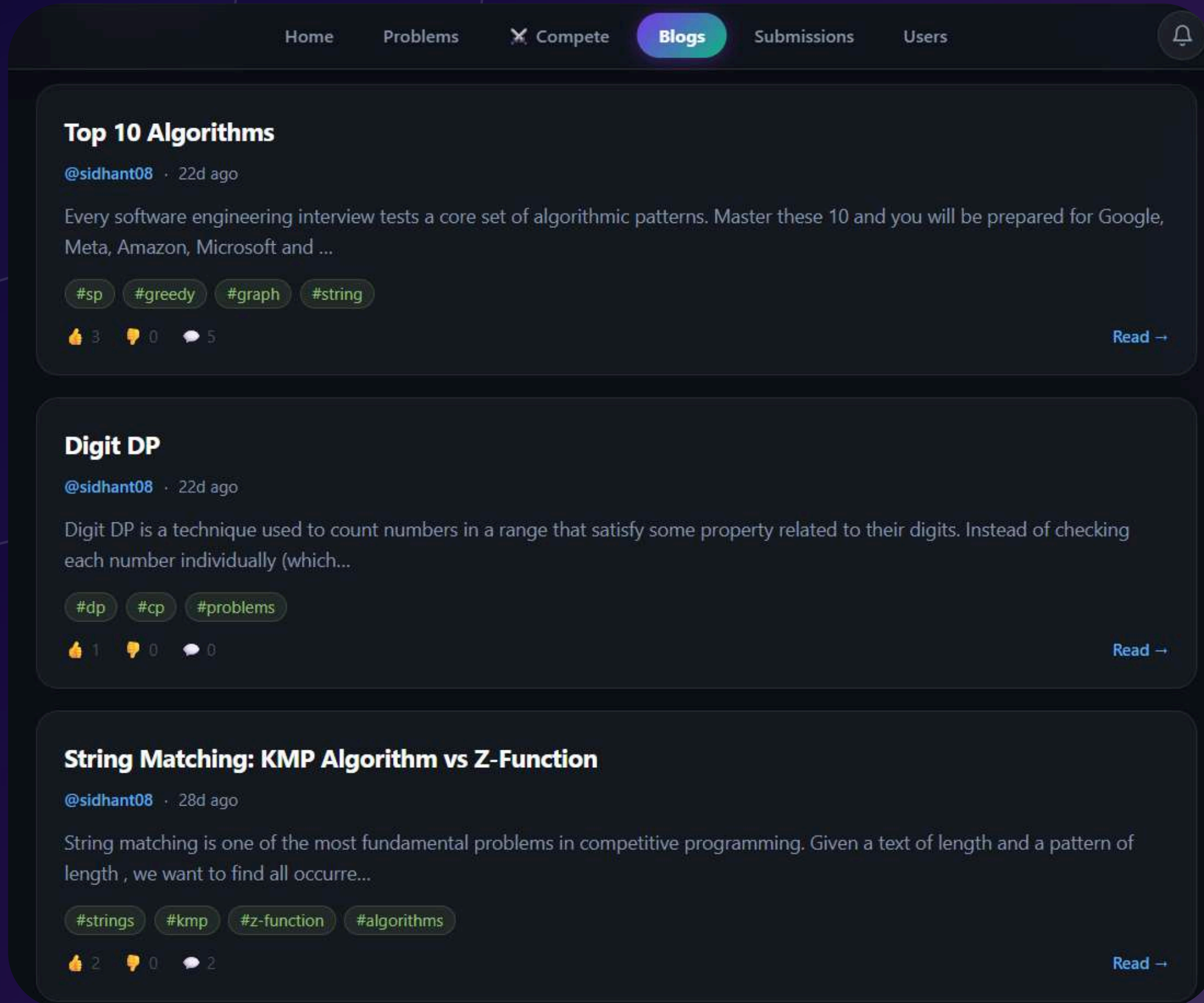
CHAT 2

cdxdx hi

sidhant08 Hello

- Live chat with opponent during battle
- Timer-based real-time coding competition

Blogs



- Blogs section in which a user can write a blog about any topic and also tag the blog.
- The tags can be used to search the blog or can be searched by matching the content of blog
- The blog can be upvoted or downvoted by other users.

Blogs

Blog

_id	ObjectId	Primary Key
title	String	Required, max 200 chars
content	String	Required
authorHandle	String	Indexed
tags	String	Lowercase, indexed
likes	String	Array of user handles
dislikes	String	Array of user handles
createdAt	Date	Default: now
updatedAt	Date	Default: now

Indexes: authorHandle + createdAt, createdAt, tags

Comment

_id	ObjectId	Primary Key
blogId	ObjectId	FK → Blog (cascade delete)
authorHandle	String	Required
content	String	Required, max 2000 chars
createdAt	Date	Default: now

Index: blogId + createdAt

Messaging

CODEJUDGE

Messages

2 conversations

Search conversations...



Sidhant Nirupam

hmm 🤔

Apr 22

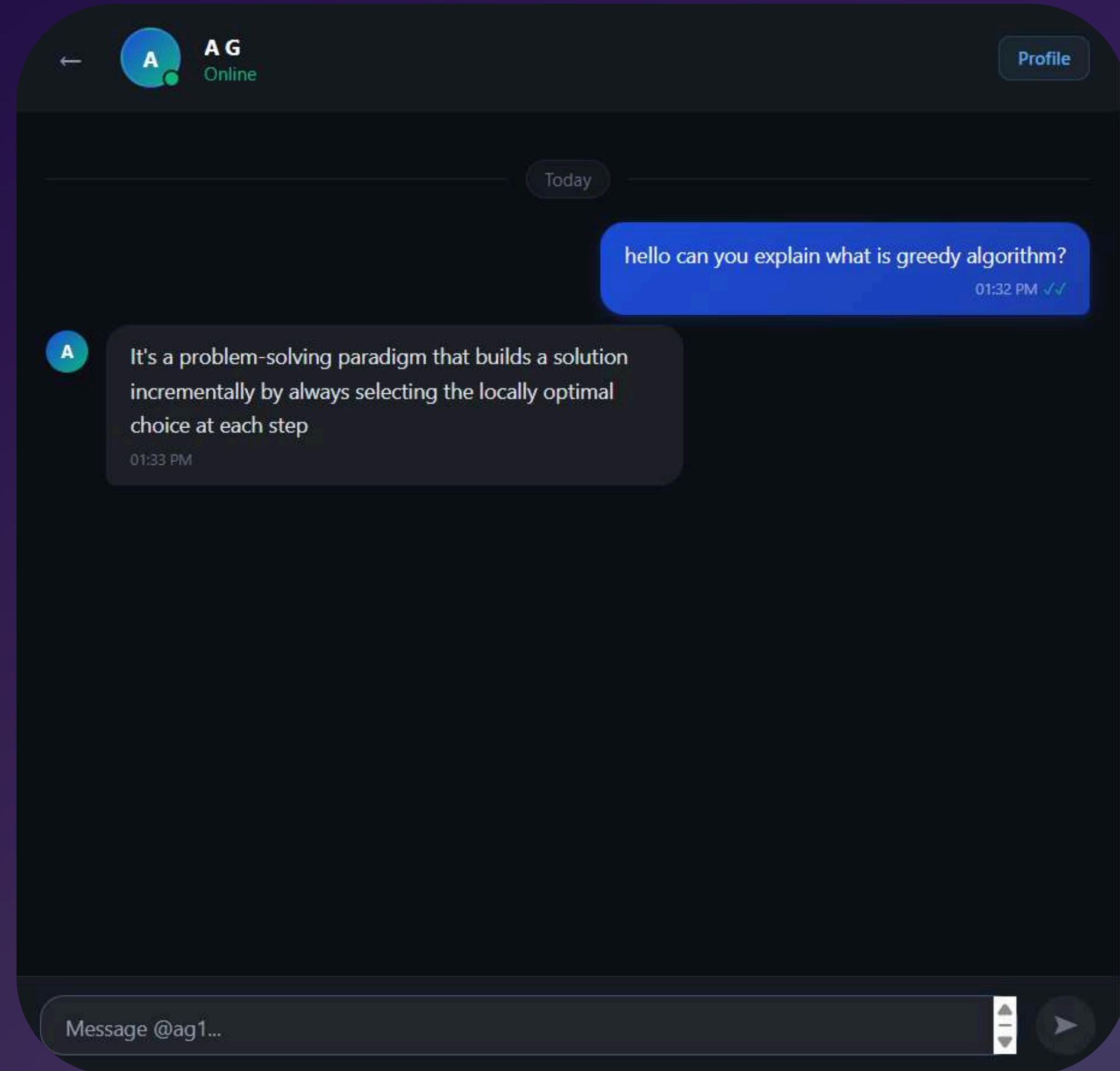


cdx dx

You: Test message

Apr 22

- Find users from user list and message them
- Ask queries or just talk



Messaging

Message

_id	ObjectId	Primary Key
sender	String	User handle, indexed
receiver	String	User handle, indexed
content	String	Required, max 2000 chars
isRead	Boolean	Default: false
timestamp	Date	Default: now, indexed

Index: sender + receiver + timestamp (compound)

User profile

- View user details (name, handle, join date)
- Rating system with max rating and badge (e.g., Platinum)
- Track battle stats (wins, losses, win rate)
- Friends list and social connections
- Performance overview in a clean dashboard
- User settings for profile management

Sidhant Nirupam

@sidhant08

★ Rating: **1315** (max: 1345 · **Platinum**)

🛡️ Badge:  **Platinum**

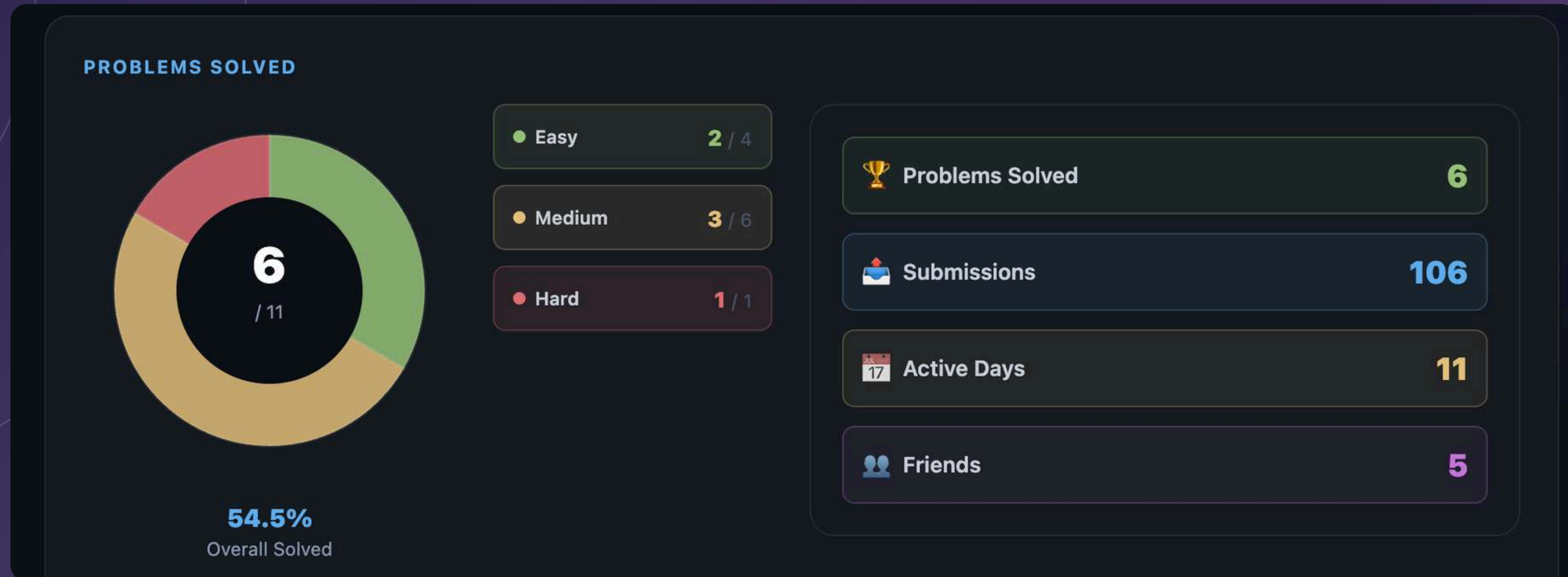
✂️ Battles: **26W** / **4L** · **87% WR** (30 games)

📅  17 Joined 2026-03-23

👤 My Friends (5)

⚙️ [Settings](#)

User analytics



- Problems Solved Overview: Summarizes total problems solved along with a breakdown using easy, medium, hard tags to reflect difficulty distribution.
- Activity Insight: Highlights consistency through metrics like active days and overall submission activity.
- Performance Distribution: Combines solved counts and difficulty tags to give a balanced view of progress and skill level.

SUBMISSION ACTIVITY



Problems Solved (All Time)

6



Problems Solved (Last Year)

6



Problems Solved (Last Month)

5



Best Streak (All Time)

2 days



Best Streak (Last Year)

2 days

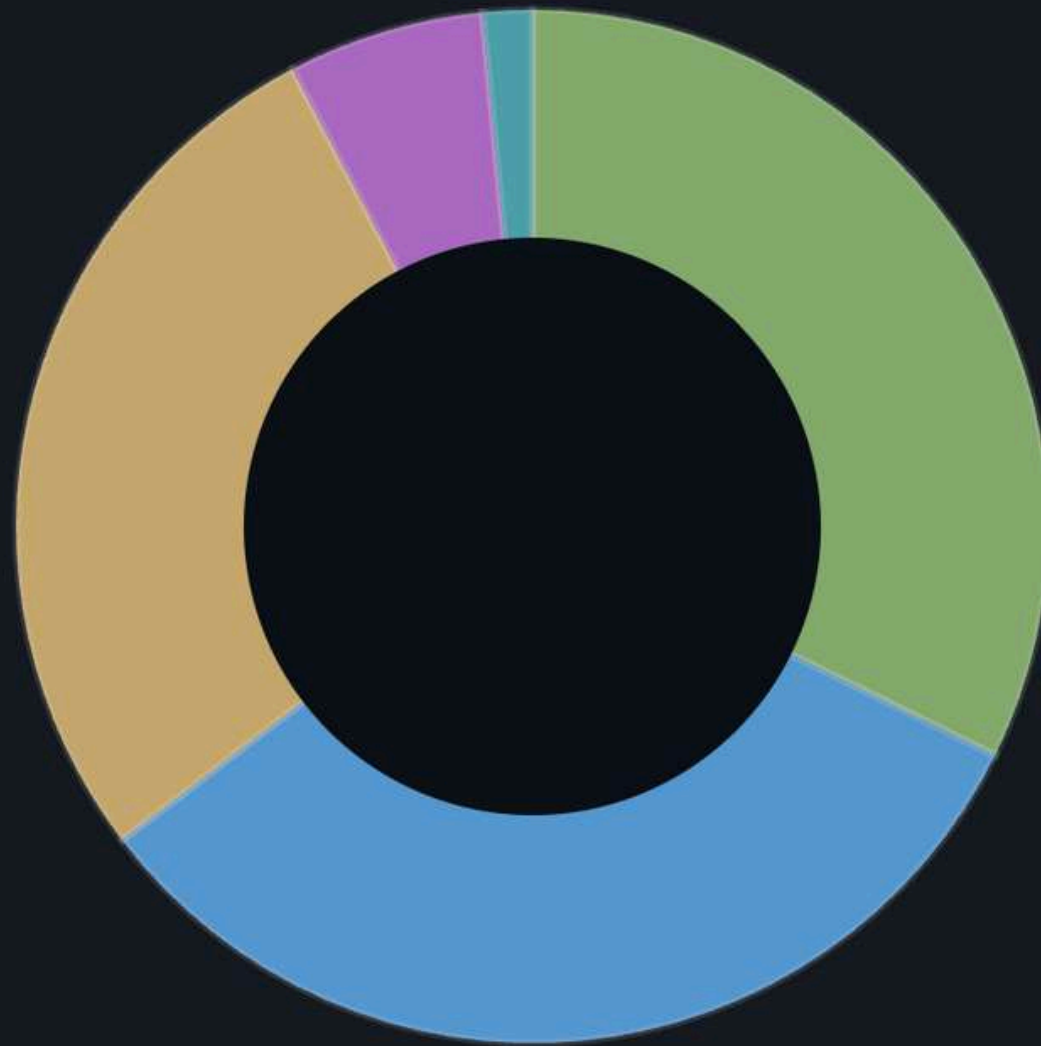


Best Streak (Last Month)

2 days

- Submission activity heatmap for daily consistency tracking
- Problems solved stats (All time, Yearly, Monthly)
- Track best solving streaks over time
- Performance analytics in an intuitive dashboard

TAG DISTRIBUTION



● strings 21

● implementation 21

● arrays 18

● dp 4

● greedy 1

- Pie Chart Overview: A visual breakdown of problem tags (like greedy, strings, dp) showing how your solved problems are divided across topics.
- Topic Focus: Makes it easy to see which areas you spend most of your time on at a glance.
- Balance of Practice: Helps you understand whether your problem-solving is well-rounded or concentrated in a few tags.

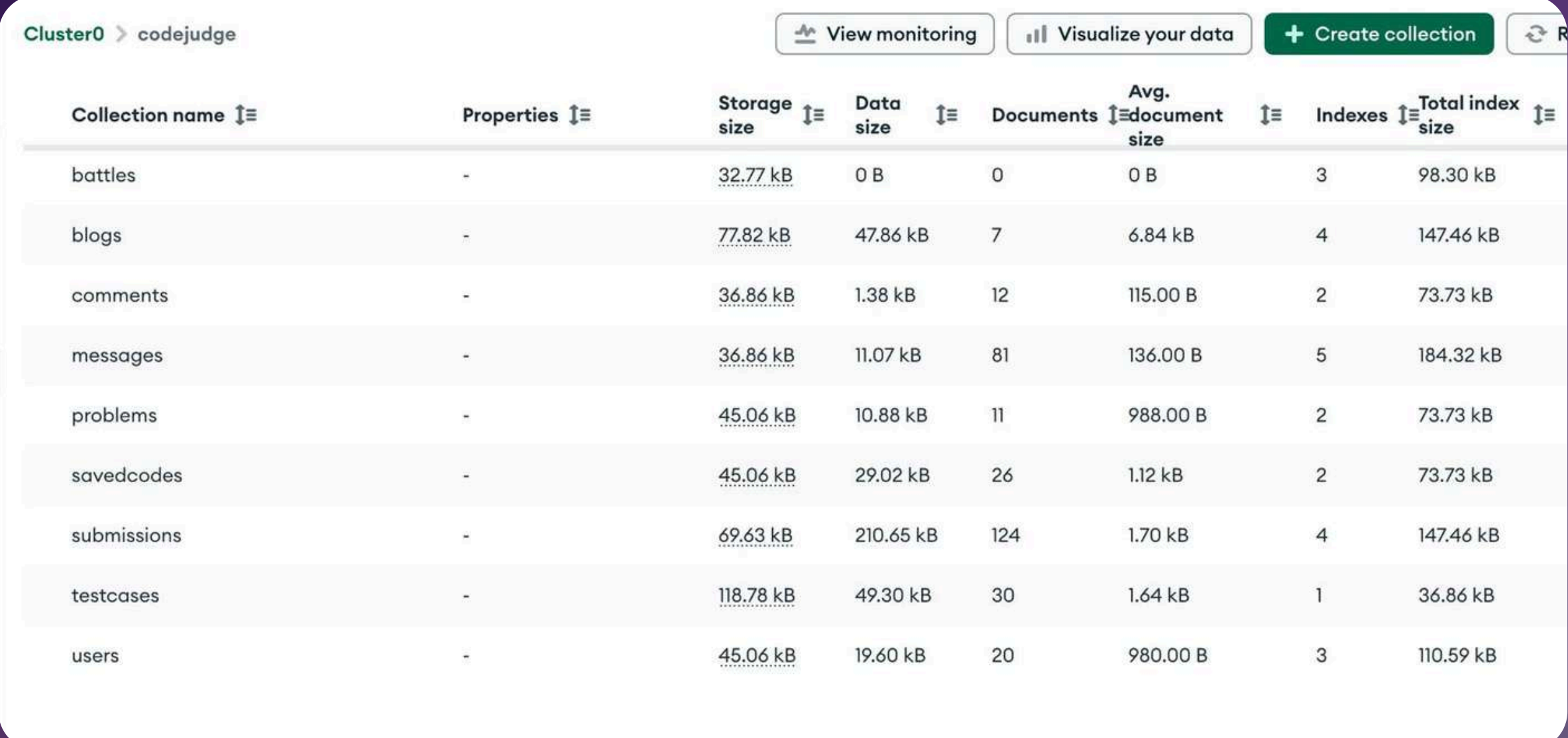
User

userhandle	String	Unique, required
firstName	String	
lastName	String	
email	String	Unique, required
imgPath	String	Profile picture path
role	String	"user" or "admin"
rating	Number	Default: 0
badge	Object	
└ name	String	e.g. "Bronze"
└ emoji	String	e.g. "🏆"
└ color	String	Hex color
└ imgPath	String	Badge image path
Friends	[String]	Array of user handles

System Database Overview

→ Collections : battles, blogs, comments, messages, problems, savedcodes, submissions, testcases, users

→ Centralized MongoDB database for structured data management



Cluster0 > codejudge

[View monitoring](#) [Visualize your data](#) [+ Create collection](#) [Refresh](#)

Collection name	Properties	Storage size	Data size	Documents	Avg. document size	Indexes	Total index size
battles	-	32.77 kB	0 B	0	0 B	3	98.30 kB
blogs	-	77.82 kB	47.86 kB	7	6.84 kB	4	147.46 kB
comments	-	36.86 kB	1.38 kB	12	115.00 B	2	73.73 kB
messages	-	36.86 kB	11.07 kB	81	136.00 B	5	184.32 kB
problems	-	45.06 kB	10.88 kB	11	988.00 B	2	73.73 kB
savedcodes	-	45.06 kB	29.02 kB	26	1.12 kB	2	73.73 kB
submissions	-	69.63 kB	210.65 kB	124	1.70 kB	4	147.46 kB
testcases	-	118.78 kB	49.30 kB	30	1.64 kB	1	36.86 kB
users	-	45.06 kB	19.60 kB	20	980.00 B	3	110.59 kB

Thank You